

QuantERA Proposal Template

Project Acronym

PA-QML

Project Title

Physics-Aware Quantum Machine Learning: From Distributed Hardware to Certified Intelligence

Topic:



Quantum Phenomena and Resources



Applied Quantum Science

Coordinator contact point for the proposal

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People involved in the realisation of the project (**please enter these names and their email addresses in the financial template as well**)

Partner Number	Country	Institution/ Department	Name of the Principal Investigator (PI)¹	Name of the co-Investigators²	Name of the other personnel participating in the project³
1 <i>Coordinator</i>	South Korea	Seoul National University / Department of Psychology; Department of Brain and Cognitive Sciences; Interdisciplinary Program in Artificial Intelligence	Jiook Cha		Junghoon Justin Park; Maria Pak; Seung Yun Choi; Eunji Lee; Bo-Gyeom Kim; Heehwan Wang; Sangyoon Bae; Jungwoo Seo; Seungju Lee; Jubin Choi; Danny

¹ The Principal Investigator (PI) is the point of contact of the partner for the corresponding Research Funding Organisation.

² A co-investigator is a known scientist and/or group leader making a substantial contribution to the project.

³ If the name is for the moment unknown, specify the level of expertise sought (PhD, post-doc, engineer, professor...).

					Dongyeop Han; Kyungjin Oh; Ahhyun Lucy Lee; Seok Jin Moon; Sebin Lee; Yujin Yun; Allison Eun Se You; Bohee Lee
2	South Korea	Yonsei University / Department of Physics	Hwidong Yoo		Sungwon Kim; Yun Eo; Haeun Jang
3	Germany	Fraunhofer Institute for Cognitive Systems IKS / Department "Quantum-enhanced AI"	Jeanette Miriam Lorenz		Alona Sakhnenko
4	Italy	University of Naples Federico II	Giovanni Acampora	Roberto Schiattarella	Autilia Vitiello; Angela Chiatto

Duration: 36 months

Summary of the project:

Physics-Aware Quantum Machine Learning (PA-QML) establishes foundational frameworks for scalable quantum computing through multi-chip coordination protocols and reliability-certified quantum algorithms.

Challenge: Quantum Machine Learning (QML) promises exponential advantages for high-complexity data, yet four fundamental barriers prevent practical deployment: (1) **scalability**—NISQ devices cannot process high-dimensional scientific data; (2) **trainability**—gradient-based methods suffer barren plateaus, while gradient-free alternatives converge slowly with high sampling overhead, and expressive circuits often remain classically simulable; (3) **temporal expressibility**—existing QML architectures lack mechanisms for long-range sequence modeling; (4) **reliability**—no certification standards exist for QML robustness against adversarial attacks or hardware noise.

Innovation: We pioneer four fundamental breakthroughs: (1) **Improved Scalability via Multi-Chip Ensembles**—partitioning high-dimensional data across independent QPUs with selective quantum connections; (2) **Enhanced Trainability via Quantum Forward-Forward with Hybrid Quantum Genetic Algorithms**—combining local "goodness" objectives with quantum evolutionary search to train deep, non-simulable circuits without gradients; (3) **Temporal Expressibility via Quantum State Space Models** with three-branch quantum superposition; (4) **Reliability via Fuzzy Quantum Diffusion**—exploiting physical quantum noise as a generative resource for hardware-native generative modeling.

Enhanced Validation Strategy: Rather than claiming immediate quantum supremacy, we establish validated theoretical frameworks through staged validation with rigorous classical baselines ensuring objective quantum advantage assessment.

Consortium: Our international partnership combines Seoul National University's multi-chip ensemble expertise, University of Naples' quantum computational intelligence leadership, Fraunhofer IKS's reliability and robustness standards, and Yonsei University's CERN collaboration experience.

Expected Impact: This project transforms QML from experimental demonstrations to preliminary engineering validation, establishing Europe-Asia leadership in practical quantum machine learning while training the next generation of quantum algorithm engineers for industry deployment.

Relevance to the topic addressed in the call:

Novel quantum algorithms and software stacks: Four algorithms—Multi-Chip Ensembles, Quantum Forward-Forward with Hybrid Quantum Genetic Algorithms (QFF-HQGA), Quantum State Space Model (Q-SSM), Fuzzy-Quantum Diffusion—target scalability, trainability, temporal expressibility, and reliability bottlenecks, integrated into an open-source library.

New architectures and programming paradigms, including hybrid approaches: Multi-Chip Ensembles introduce distributed quantum computing via QPU aggregation. All innovations are hybrid quantum-classical architectures optimized for NISQ constraints.

Development of noisy intermediate-scale quantum platforms: Fuzzy-Quantum Diffusion exploits noise as a generative resource. The QML Reliability Certification Standard provides the first formal robustness metrics.

Resource theory for quantum information: We establish the first resource theory for distributed QML, proving ensemble aggregation achieves collective advantage while Selective Entanglement optimizes expressivity-trainability trade-offs.

Superposition, interference, and entanglement for enhanced functionalities: Q-SSM uses three-branch superposition for interference-like features; Multi-Chip Ensembles

leverage restricted entanglement to prevent barren plateaus; QFF-HQGA employs entangled crossover for evolutionary optimization.

Understanding and control of open quantum systems: Fuzzy-Quantum Diffusion models decoherence as continuous uncertainty via POVMs, converting noise into a computational resource.

1. Excellence (max. 6 pages)

1.1 Targeted Breakthrough, Baseline of Knowledge and Skills

Targeted Breakthroughs

This project targets **four fundamental barriers** preventing Quantum Machine Learning (QML) from achieving practical advantage. Current QML remains confined to small-scale demonstrations reliant on heuristic ansatzes that become untrainable at scale. We shift the paradigm from "heuristic QML" to "**Physics-Aware & Scalable QML**"—algorithms designed for NISQ constraints **with rigorous theoretical foundations and practical validation pathways**.

Breakthrough 1 – Scalability via Multi-Chip Ensembles: We partition high-dimensional data across k independent QPUs, each processing locally-correlated inputs [1]. **Selective Entanglement** introduces inter-chip quantum connections only for globally-dependent features. Classical aggregation provides algorithmic error mitigation: independent chip errors average out (variance reduction $\propto 1/k$) without resource-heavy correction codes, establishing a **resource theory for "Collective Quantum Advantage"**. This architecture aligns with emerging modular quantum hardware platforms (e.g., IBM Quantum, IonQ, Pasqal), enabling direct deployment as interconnected QPU systems become available.

Breakthrough 2 – Trainability via Quantum Forward-Forward & Hybrid Quantum Genetic Algorithm: We integrate **Quantum Forward-Forward (QFF)** [2] with **Hybrid Quantum Genetic Algorithm (HQGA)** [3]. QFF decomposes deep circuits into layers optimizing local goodness objectives, mathematically circumventing barren plateaus [4] while preserving expressivity. Hybrid Quantum Genetic Algorithm encodes parameters as **quantum chromosomes**—superpositions enabling parallel fitness evaluation—using **entangled crossover** and **quantum elitism**.

Breakthrough 3 – Temporal Expressibility via Q-SSM: We develop **Quantum State Space Models (Q-SSM)** [5] integrating three-branch quantum superposition with LSTM-style gating. Quantum circuits access 2^n -dimensional Hilbert space with $O(n \times l)$ parameters versus $O(N^2)$ classically. The architecture achieves **$O(L)$ complexity** through chunked processing, matching classical state space models [6], [7] efficiency while exploiting quantum feature expressivity.

Breakthrough 4 – Reliability via Fuzzy Quantum Diffusion: We exploit physical quantum noise as a generative resource [8], [9]. **Fuzzy Quantum Logic** via Positive Operator-Valued Measures [10] provides continuous measurements, enabling **Hardware-Attention U-Net** to learn device-specific noise fingerprints. To curb uncertainty from noisy models further, we design **robustness and safety certification** [11], [12] that meet **industrial-grade robustness** [13] standards, with verified resilience to targeted attacks and untargeted perturbations.

Consortium Expertise:

Partner	Expertise	Project Role
SNU (Cha)	Multi-Chip Ensembles, Quantum State Space Models, Quantum Transformers	Coordinator; Architecture Lead (WP1, WP2)
Naples (Acampora)	Evolutionary Algorithms, Fuzzy Logic, Quantum Computational Intelligence	Optimization & Generative Model Lead (WP3)

Partner	Expertise	Project Role
Fraunhofer IKS (Lorenz)	QML, robustness certification, European QC Benchmarking Committee	Certification Lead (WP5, WP6)
Yonsei (Yoo)	CMS/CERN collaboration, Q-ABCDiCo, Temporal Convolutional Network-VQC, Quantum Vision Transformer for High Energy Physics	Domain Validation Lead (WP4)

Specific Objectives

Objective	Goal	Measure	Target	Timeline
O1: Scalability	Multi-Chip Ensembles with Selective Entanglement	Accuracy retention vs monolithic circuit	$\geq 90\%$ accuracy at $2\times$ effective qubit scaling; $< 10\%$ variance across configurations	M12→M30
O2: Trainability	Quantum Forward-Forward & Hybrid Quantum Genetic Algorithm Optimization	Convergence in barren plateau regimes; measurement count	Convergence at > 10 layers where BP fails; $\geq 50\%$ fewer circuit evaluations vs parameter-shift	M12→M24
O3: Temporal	Quantum State Space Model with three-branch superposition	Memory capacity; complexity scaling	$\geq 1.5\times$ memory capacity vs classical SSM; verified $O(L)$ scaling	M18→M30
O4: Reliability	Fuzzy Quantum Diffusion & Safety Certification	FID under noise; robustness against perturbations	$FID \leq 1.5\times$ classical baseline under 10^{-3} gate error; Performance degradation $\leq 10\%$	M18→M36
O5: Validation	Domain science validation of O1-O4	HEP/Neuro/Cyber benchmarks	Statistical quantum advantage ($p < 0.05$) in ≥ 1 domain metric	M6→M36

O5 Validation Details:

- **HEP:** Multi-Chip Quantum Vision Transformer for calorimeter classification; Q-ABCDiCo ($\sim 1,372$ parameters) with Distance Correlation penalty for background estimation; Temporal Convolutional Network-Variational Quantum Circuits with Noise2Noise self-supervision for waveform denoising
- **Neuroscience:** Quantum State Space Models and Quantum Transformers for EEG/fMRI spatio-temporal analysis (→O3); Quantum Forward-Forward & Hybrid Quantum Genetic Algorithm optimized QAOA for brain network modularity optimization and VQE for active inference modeling (→O2)
- **Cybersecurity:** Fuzzy Quantum Diffusion for anomaly detection via hardware fingerprinting; certified robustness against targeted (e.g. adversarial attacks) and untargeted (e.g. input or device noise) perturbations.

1.2 Novelty, Level of Ambition and Foundational Character

Advance Beyond State-of-the-Art

Challenge	SOTA Limitation	Our Advance
Scalability	Circuit cutting incurs exponential sampling overhead; no multi-modal fusion theory exists	Resource theory for Collective Quantum Advantage [1]: ensemble aggregation preserves expressivity; partitioning eliminates inter-chip entanglement costs; Selective Entanglement optimizes expressivity-trainability trade-off
Trainability	Gradient-based: barren plateaus at $O(n)$ depth [4]. Gradient-free: slow convergence, high sampling. Trainable circuits often classically simulable	Symbiotic Quantum Forward Forward & Hybrid Quantum Genetic Algorithm : local goodness objectives [2] mathematically avoid barren plateaus; quantum chromosomes with entangled crossover [3] navigate non-simulable deep circuit landscapes without gradient computation
Temporal	Quantum RNNs collapse sequences via pooling; no quantum state space model exists	Novel Quantum State Space Model architecture [5]: measurement-based superposition ($\alpha, \beta, \gamma \in \mathbb{C}$) preserves gradient flow; LSTM gating enables selective memory; chunked processing achieves $O(L)$ complexity matching classical state space model [6]
Reliability	Noise treated as error requiring resource-heavy correction; no QML certification standards	Noise-as-resource paradigm for QML [8]: Fuzzy Positive Operator-Valued Measures [10] enable hardware-native generative modeling; safety certification protocol [11] with formal guarantees.

Level of Ambition

Foundational, not incremental: We establish new theoretical frameworks—resource theory for distributed QML, mathematical proof of barren plateau avoidance via local objectives, formal certification metrics—rather than algorithmic refinements.

"Big Science" Validation: We validate on datasets with genuine complexity, not toy benchmarks:

- **HEP (LHC/CMS):** Petabyte-scale data with complex non-local correlations. Q-ABCDiCo achieves background estimation with $\sim 1,372$ parameters (vs $\sim 10^5$ classical). Quantum Vision Transformer targets merged electron classification at $\Delta R \approx 0.001$ resolution.
- **Neuroscience:** High-dimensional, data-scarce clinical signals [14]. Quantum State Space Model captures long-range temporal patterns in EEG/fMRI. Quantum Forward-Forward & Hybrid Quantum Genetic Algorithm enables training of QAOA for brain network modularity optimization and VQE for active inference modelling [15].
- **Cybersecurity:** Dynamic threat landscapes requiring certified robustness. Safety certification provides formal adversarial guarantees unprecedented in QML [16].

Industrial Certification Standard: we transform QML from experimental demonstrations to certifiable engineering—a prerequisite for safety-critical deployment in automotive, medical, and financial sectors.

1.3 Concept and Methodology

Overall Concept: Physics-Aware QML Stack

We assume NISQ devices remain fragmented and noisy. Rather than waiting for fault tolerance, we redesign algorithms to exploit these constraints.

Core hypothesis: Quantum advantage lies in *feature representation* (Hilbert space expressivity with parameter efficiency), not error-free computation. Our hybrid architectures extract quantum features via shallow, noise-resilient circuits while delegating temporal memory and output generation to classical components proven effective for these tasks.

Methodology

M1: Multi-Chip Ensembles (→O1: Scalability)

Mechanism: Partition input $x \in \mathbb{R}^d$ into k subsets $\{x_1, \dots, x_k\}$ assigned to independent n -qubit chips. Each chip i applies modality-specific ansatz $U_i(\theta_i)$. **Selective Entanglement**

introduces controlled inter-chip operations only between features with high classical correlation (identified via mutual information analysis). Outputs aggregate via learned weights $\hat{y} = \sum_i w_i \cdot f(\langle \psi_i | M | \psi_i \rangle)$.

Scientific basis: Partitioning restricts entanglement entropy growth across the full system, provably mitigating barren plateaus [4] (concentration of measure scales with entangled subsystem size, not total qubits). Independent chip errors are uncorrelated; aggregation reduces variance by factor $1/k$ without correction overhead [1].

M2: QFF-HQGA Optimization (→O2: Trainability)

Mechanism: Decompose L -layer circuit into L independent optimization problems. Layer l minimizes local goodness $G_l(\theta_l) = \langle \psi_{pos} | M_l | \psi_{pos} \rangle - \langle \psi_{neg} | M_l | \psi_{neg} \rangle$, where ψ_{pos}/ψ_{neg} are states from positive (real) and negative (corrupted) data passes. Within each layer, HQGA encodes parameters as quantum chromosomes $|\theta\rangle = \sum_j \alpha_j |\theta_j\rangle$. **Entangled crossover**

(controlled rotations correlating successful parameter traits) and **quantum elitism** (amplitude amplification of high-fitness states) evolve the population.

Scientific basis: Local objectives have $O(1)$ gradient variance—versus exponentially vanishing global gradients [4]—because each layer's loss landscape is shallow [3]. Quantum parallelism evaluates population fitness in $O(1)$ circuit executions per generation, eliminating the measurement overhead of parameter-shift gradient estimation.

M3: Q-SSM Architecture (→O3: Temporal Expressibility)

Mechanism: For sequence $\{x_t\}$, partition into chunks. Each chunk passes through three parallel VQCs producing measurements $\{m_\alpha, m_\beta, m_\gamma\}$. Combine via trainable complex coefficients: $h_q = \alpha \cdot m_\alpha + \beta \cdot m_\beta + \gamma \cdot m_\gamma$, where $\alpha, \beta, \gamma \in \mathbb{C}$

This **measurement-based superposition** enables interference-like feature combination while maintaining gradient flow. Classical LSTM gates process temporal dynamics:

$$c_t = f_t \odot c_{t-1} + i_t \odot h_q, o_t = \tanh(c_t) \odot \sigma(W_o \cdot h_q)$$

Chunked processing divides long sequences into manageable segments, with LSTM gates maintaining state across chunks—achieving $O(L)$ complexity analogous to classical state space models [6], [7] while leveraging quantum Hilbert space expressivity.

Scientific basis: Quantum circuits provide 2^n -dimensional feature space with $O(n \times l)$ parameters versus $O(N^2)$ for equivalent classical layers. LSTM gating—proven optimal for temporal credit assignment—handles sequence memory. Complexity is $O(L)$ in sequence length (constant quantum cost per chunk).

M4: Fuzzy Quantum Diffusion + Reliability & Safety Certification (→O4: Reliability)

Mechanism: Forward diffusion uses physical noise channels (amplitude damping, dephasing, crosstalk) instead of synthetic Gaussian noise. **Positive Operator-Valued Measures (POVMs)** provide continuous "fuzzy" measurements $x_t = \text{Tr}[E_t(\rho) \cdot E_{\text{detect}}]$ where E_t parameterizes measurement sharpness by decoherence time t .

Hardware-Attention U-Net learns device-specific reversal using calibration maps (T_1 , T_2 , readout error rates). **Robustness validation** requires empirically stress testing the models under various perturbing conditions to ensure that its output remains within certain bounds, which can be captured through e.g. **Lipschitz constant:** $L = \max_{\{x, x'\}} \frac{\|f(x) - f(x')\|}{\|x - x'\|}$.

Scientific basis: Physical noise is deterministic given device characterization—unlike stochastic Gaussian—enabling precise reversal learning [8]. Fuzzy measurements [10] preserve information lost in sharp projection, maintaining gradient flow for hybrid training. Lipschitz bounds [11] provide formal guarantees: output perturbation $\leq L \times$ input perturbation.

M5: Domain Validation (→O5)

Domain	Task	Method	Quantum Advantage Hypothesis
HEP	Background estimation	Q-ABCDiSCo + DiSCo penalty	Parameter efficiency (~1,372 vs ~10 ⁵)
	Waveform denoising	TCN-VQC (4 qubits) + Noise2Noise	Self-supervised learning without clean labels
	Merged electron ID	Multi-Chip Quantum ViT	$\Delta R \approx 0.001$ resolution via quantum attention
Neuro	EEG/fMRI analysis	Q-SSM + Quantum Transformers	Temporal expressivity (→O3)
	Brain network modularity optimization	QFF-HQGA optimized QAOA/VQE	Trainability for combinatorial optimization (→O2)
Cyber	Anomaly detection	Fuzzy Quantum Diffusion	Hardware fingerprint sensitivity
	Robustness certification	Targeted / untargeted perturbation robustness	Adversarial robustness and data privacy

Risk Mitigation

Risk	Likelihood	Mitigation Strategy
Barren plateaus persist despite QFF	Medium	Fallback to pure HQGA (entirely gradient-free); reduce circuit depth
Ensemble accuracy degradation	Medium	Weighted voting by per-chip reliability scores; adaptive chip selection
Hardware noise exceeds Fuzzy model capacity	Low	Pivot to Fuzzy Quantum GANs with stabilized training dynamics
Domain data access delays	Low	CERN Open Data Portal; PhysioNet EEG; public intrusion detection datasets

1.4 Interdisciplinary Nature

Research Disciplines and Integration

Discipline	Contribution	Integration Point
Quantum Information Science	Ansatz design, entanglement theory, variational circuits	Provides computational substrate and expressivity theory
Computational Intelligence	Fuzzy Logic, Evolutionary Algorithms	Solves trainability/robustness where pure QIS methods fail
Software Engineering & Reliability	benchmarking, certification protocols	Transforms theory into deployable, verifiable standards
Domain Sciences (HEP, Neuroscience)	Complex real-world datasets and domain constraints	Provides validation complexity; grounds quantum advantage claims

Cross-Fertilization: Annual methodology exchanges ensure knowledge transfer (Naples: Fuzzy/evolutionary methods; SNU: Q-SSM/Multi-Chip protocols; Fraunhofer: Safety certification). Domain Challenge Sprints apply theoretical advances to real problems—HEP jet classification, EEG/fMRI temporal modelling, brain network modularity optimization, adversarial robustness testing—evaluated by domain-specific metrics.

Interdisciplinary Synergies:

- **Fuzzy Logic + Quantum Mechanics:** Biological uncertainty (neuroscience) and hardware noise (NISQ) unified as continuous "degrees of truth" via POVMs
- **Evolutionary Algorithms + Quantum Circuits:** QML models that *evo/ve* through non-convex landscapes where gradient-based optimization fails
- **Distributed Computing + HEP:** Multi-Chip protocols validated on particle data with intrinsic non-local correlations—ideal for demonstrating ensemble aggregation preserves quantum expressivity

A unified software architecture with standardized interfaces (Physics, Logic, Certification modules) ensures integration at the code level.

1.5 Gender Dimension and Open Science Practices

Gender Dimension

The foundational QML algorithms are gender-neutral. For the **Neuroscience validation (O5)**, where EEG/fMRI data involves human subjects, we implement **gender-stratified validation**: balanced training data (50/50 male/female) and sex as an explicit covariate to prevent algorithmic bias in brain signal analysis.

The consortium includes **female leadership** (PD Dr. habil. Jeanette Miriam Lorenz, Fraunhofer IKS) as Principal Investigator, serving as a role model for early-career researchers.

Open Science Practices

- **Open Access:** 100% of publications via Green (arXiv, Zenodo) or Gold routes.
- **FAIR Data:** Challenge datasets (anonymized EEG, LHC subsets) deposited in Zenodo with DOIs. Data Management Plan delivered by M3.
- **Open Source:** "Physics-Aware QML" library released under Apache 2.0 license on GitHub, with Jupyter notebook tutorials for reproducibility.
- **Sensitive Data:** Neuroscience data pseudonymized at source (GDPR/PIPA compliant). Cybersecurity logs screened for PII before consortium sharing.

2. Impact (max. 3 pages)

2.1 Expected Impacts

This project addresses the QuantERA QPR call objective to *"explore novel quantum phenomena and resources... and address major challenges preventing broad applications."* We establish a **resource theory for scalable and reliable QML** with the following impacts:

Quantified Impact Indicators

Impact	Objective	Indicator	Baseline	Target	Timeline
Distributed Computing	O1	Accuracy retention at 2× qubit scaling	Single-chip: ~20-50 qubits	$\geq 90\%$, variance $\propto 1/k$	M30
Trainability	O2	Convergence in barren plateau regimes	Gradient methods fail >10 layers	QFF-HQGA converges >10 layers	M24
Temporal Learning	O3	Memory capacity vs. classical SSM	Quantum RNNs collapse sequences	$\geq 1.5\times$ classical; $O(L)$ complexity	M30
Reliability	O4	Certified robustness (Lipschitz bound)	No QML certification exists	$L < 2.0$; FID $\leq 1.5\times$ at 10^{-3} error	M36

Key Contributions:

- **O1:** "Collective Quantum Advantage" via Multi-Chip Ensembles [1]—ensemble aggregation provides algorithmic error mitigation (variance $\propto 1/k$) without correction codes
- **O2:** QFF-HQGA resolves the optimization trilemma—local goodness objectives [2] avoid barren plateaus; quantum evolutionary search [3] eliminates gradient overhead
- **O3:** Q-SSM [5] achieves $O(L)$ complexity matching classical state space models [6] while exploiting 2^n -dimensional Hilbert space with $O(n \times l)$ parameters
- **O4:** Safety certification [11], [12] provides formal Lipschitz guarantees; Fuzzy POVMs [10] enable noise-as-resource paradigm [8]

Domain Science Validation (→O5)

Domain	Task	Method	Quantum Advantage Hypothesis	Success Metric
HEP	Background estimation	Q-ABCDiSCo	$\sim 1,372$ vs. $\sim 10^5$ classical parameters	$p < 0.05$
	Waveform denoising	TCN-VQC + Noise2Noise	Self-supervised without clean labels	SNR improvement
	Merged electron ID	Multi-Chip Quantum ViT	$\Delta R \approx 0.001$ resolution	Classification accuracy

Domain	Task	Method	Quantum Advantage Hypothesis	Success Metric
Neuro	EEG/fMRI analysis	Q-SSM	Long-range temporal patterns	Memory $\geq 1.5\times$ classical
	Brain network modularity optimization	QFF-HQGA QAOA/VQE	Trainability for combinatorial problems [15]	Convergence
Cyber	Anomaly detection	Fuzzy Quantum Diffusion	Hardware fingerprint sensitivity	Detection rate
	Robustness certification	Noise in QML [13]	Formal adversarial guarantees	$L < 2.0$

Transformational Societal Impact

Impact Area	Gap Addressed	Our Contribution	Societal Benefit
Industrial Certification	No QML safety/robustness standards	Pipeline-integrated certification (Lipschitz bounds, noise stability)	Enables quantum AI in safety-critical sectors (automotive, medical, finance)
Democratization	Quantum advantage restricted to elite labs	Multi-Chip Ensembles aggregate smaller QPUs	Universities/SMEs access large-scale modular QML
Privacy-Preserving Healthcare	Medical data sharing constrained by GDPR/HIPAA	Fuzzy Quantum Diffusion generates synthetic data [9]	Unlocks medical AI development without privacy compromise
Fundamental Physics	LHC analysis bottlenecks	Quantum ViT/Q-ABCDiCo with superior parameter efficiency	Accelerates discovery beyond Standard Model

2.2 Dissemination, Exploitation, and Communication

Dissemination Strategy

- **Publications (≥ 12 peer-reviewed, 100% Open Access):**

Type	Venues	Timeline	Lead
Theory	<i>Nature Comms, PRX Quantum</i>	M12-M24	SNU, All
Methods	<i>npj Quantum Info, QST</i>	M18-M30	SNU, Naples
Applications	<i>PRL, NeuroImage, IEEE S&P</i>	M24-M36	Yonsei, Fraunhofer
Tools	<i>Quantum Machine Intelligence</i>	M30-M36	Fraunhofer, Naples

- **Conferences:** IEEE QAI, IEEE QCE (2-3 papers each); NeurIPS/ICML workshops; domain conferences (CHEP, OHBM, IEEE S&P)
- **Training:** Annual consortium workshops (M12, M24, M36); Joint summer school "Practical QML for Domain Scientists" (M18)

Exploitation Plan

Innovation	IP Strategy	Pathway
Multi-Chip Ensemble Protocol	Open-source (Apache 2.0)	PennyLane/Qiskit integration
QFF-HQGA Optimizer	Open-source	Community adoption
Certification Extensions to Software Pipelines	Joint IP (Fraunhofer)	Industrial licensing
Domain Models & Datasets	Open (Zenodo, HuggingFace)	Benchmark standardization

- **Software Deliverables:** QML-Ensemble Library (GitHub); Certification Module; Domain Benchmark Suites (Q-ABCDISCo, Q-SSM-EEG, Fuzzy-QIDS)
- **Commercialization:** Hardware vendors (IBM, IQM, Pasqal)—Multi-Chip protocols; Pharma/Biotech—synthetic data licensing; Automotive/Aerospace—Fraunhofer certification consulting
- **Standardization:** IEEE P7131 (Quantum Performance Metrics); QED-C benchmarking; ISO/IEC JTC 1/SC 42 (AI certification)

Communication Activities

Audience	Channels	KPIs
Scientific	Publications, GitHub, conferences	500+ GitHub stars, 100+ forks
Industry	Workshops, white papers, advisory board	200+ webinar registrations
Policymakers	Policy briefs (M18, M36)	50+ decision-makers reached
Public	Website, social media, video series	5,000+ visitors, 10,000+ views

Data Management (FAIR Principles)

Data Type	Repository	Access	Timeline
Benchmark datasets	Zenodo (DOI)	Open	M24
Trained models	HuggingFace	Open	M30
Source code	GitHub	Apache 2.0	Continuous
Raw data	Institutional	Restricted	M36

Reproducibility: Docker containers, Jupyter notebooks, OSF pre-registration for validation experiments.

3. Implementation

3.1 Work Plan (max. 2 pages)

Overall Structure

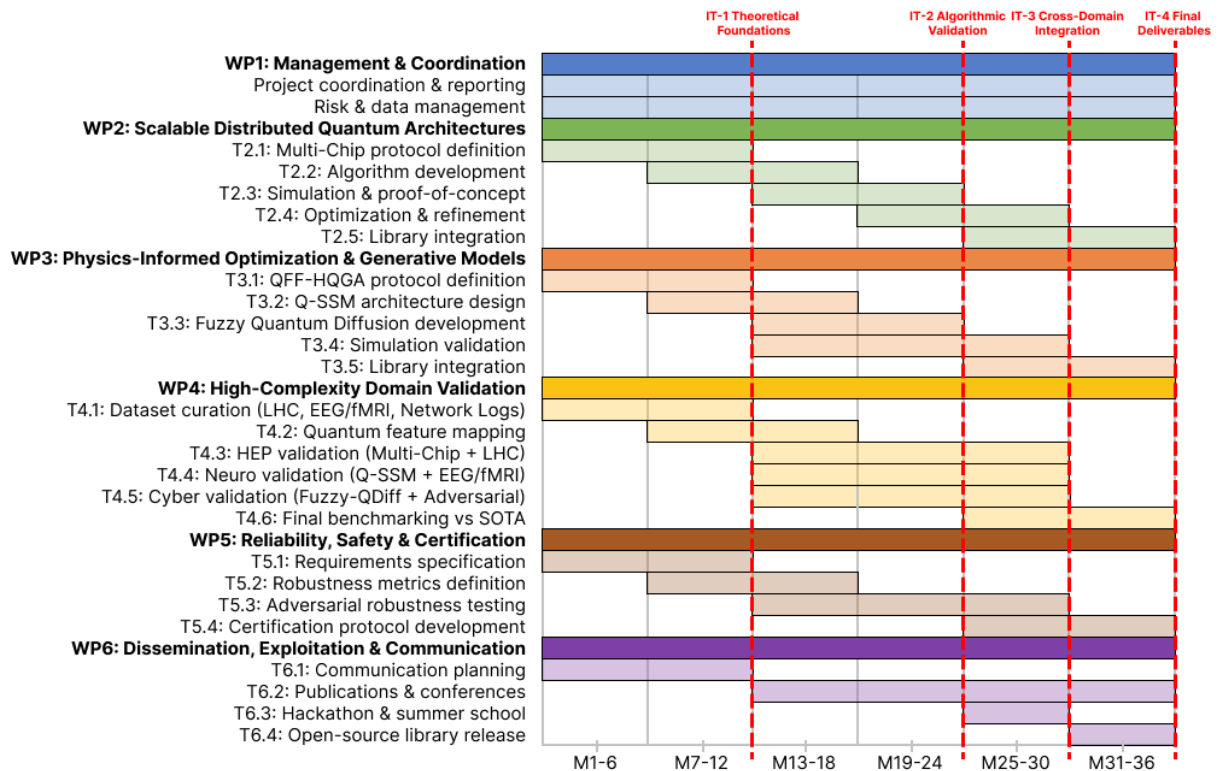
The project is organized into **six interconnected Work Packages (WPs)** creating a feedback loop between foundational theory development and empirical validation:

WP	Title	Lead	Focus
WP1	Management & Coordination	SNU	Scientific alignment, reporting, risk/data management
WP2	Scalable Distributed Quantum Architectures	SNU	Multi-Chip Ensemble protocols (→O1)
WP3	Physics-Informed Optimization & Generative Models	Naples	QFF-HQGA, Q-SSM, and Fuzzy Quantum Diffusion (→O2, O3, O4)
WP4	High-Complexity Domain Validation	Yonsei	HEP, Neuroscience, and Cybersecurity benchmarking (→O5)
WP5	Reliability, Safety & Certification	Fraunhofer	Robustness certification (→O4)
WP6	Dissemination, Exploitation & Communication	Fraunhofer	Open Science, software release, outreach

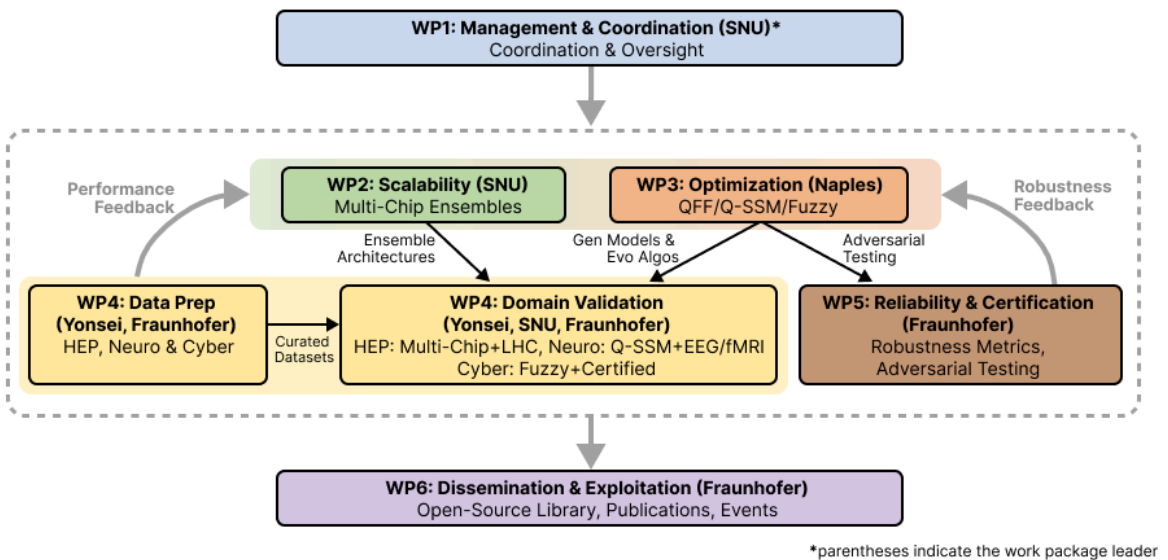
Intermediate Targets

Target	Month	Description	Success Criteria
IT-1	M12	Theoretical Foundations Established	Multi-Chip protocol specs complete; Q-SSM architecture defined; Fuzzy Logic Controllers specified; Domain datasets preprocessed
IT-2	M24	Algorithmic Validation (Simulation)	QFF-HQGA converges on >10-layer circuits where backpropagation fails; Q-SSM achieves $\geq 1.5\times$ classical memory capacity; Fuzzy-Diffusion achieves FID $\leq 1.5\times$ classical diffusion baseline
IT-3	M30	Cross-Domain Integration	Multi-Chip Ensemble reconstructs LHC particle correlations; Cybersecurity models pass robustness certification
IT-4	M36	Final Deliverables	Open-source "Physics-Aware QML" library released; Final benchmarks published

Gantt Chart: Work Package Timeline



PERT Diagram: Work Package Dependencies



Key Dependencies:

- **WP2 → WP4:** Multi-Chip Ensemble protocols feed into HEP scalability validation
- **WP3 → WP4:** Q-SSM feeds into Neuro spatio-temporal analysis; QFF-HQGA feeds into brain network modularity optimization; Fuzzy-Diffusion feeds into Cyber anomaly detection
- **WP3 → WP5:** Evolutionary QML models undergo adversarial robustness testing
- **WP4/WP5 → WP2/WP3:** Validation feedback refines foundational algorithms
- **All → WP6:** Validated results flow to dissemination

3.2 Work Packages

WP 1	Project Management & Coordination				Start Month: 1	End Month: 36
Contribution of project partners						
Partner number ⁴		1. SNU	2. Yonsei	3. Fraunhofer	4. Naples	
Total effort per partner (person*months)		9	1	2	1	
Aim of the WP						
To ensure efficient administrative and scientific coordination, timely reporting to national funding agencies and the QuantERA secretariat, and effective risk/data management across the consortium.						
Tasks						
T1.1		Consortium Coordination & Monitoring (M1–M36; SNU; All) ⁵ Day-to-day management of the project; organization of bi-annual consortium meetings (virtual/physical); monitoring of milestones and KPIs; preparation of periodic progress reports.				
T1.2		Risk & Innovation Management (M1–M36; SNU; Fraunhofer) Continuous monitoring of scientific and technical risks; execution of mitigation plans; management of Intellectual Property Rights (IPR); coordination of innovation activities.				
T1.3		Data Management & Ethics (M1–M36; SNU; Fraunhofer) Implementation of the Data Management Plan (DMP) ensuring FAIR principles; SNU oversees ethical compliance for EEG/fMRI data (Neuroscience) including IRB protocols and GDPR/PIPA compliance; Fraunhofer handles PII screening for Cybersecurity network logs.				
Deliverable		Month of delivery	Title of deliverable			
D1.1		M3	Data Management Plan			
D1.2		M18	Mid-term Progress Report			
D1.3		M36	Final Project Report			

⁴ **Bold** the partner number of the work package leader

⁵ For instance: T1.1 Development of something (M3-M6; responsible: 3; involved: 1, 4)

WP 2	Scalable Distributed Quantum Architectures				Start Month: 1	End Month: 36
Contribution of project partners						
Partner number		1. SNU	2. Yonsei	3. Fraunhofer	4. Naples	
Total effort per partner (person*months)		60	6	4	9	
Aim of the WP						
To develop theoretical protocols for scalable modular quantum computing by implementing Multi-Chip Ensembles, enabling scalable QML on fragmented NISQ hardware via parallel model aggregation. This WP directly addresses Objective 1 (Scalability) .						
Tasks						
T2.1	Multi-Chip Protocol Definition (M1–M12: SNU; Naples, Fraunhofer) Mathematical formulation of ensemble learning strategies for distributed QPUs; development of aggregation logic (voting/averaging) to combine outputs from independent quantum models; definition of inter-chip communication protocols.					
T2.2	Algorithm Development (M7–M18; SNU; Naples, Fraunhofer) Design of scalable QML architectures (Quantum Transformers, HQTCN) optimized for ensemble deployment; development of Multi-Modal Data Fusion techniques for heterogeneous data (e.g., sMRI on Chip A, fMRI on Chip B).					
T2.3	Simulation & Proof-of-Concept (M13–M24; SNU; Naples, Fraunhofer) Implementation of Multi-Chip protocols in simulation environments; validation of scalability claims on >20 qubit distributed systems; verification of barren plateau mitigation through restricted global entanglement.					
T2.4	Optimization & Refinement (M19–M30; SNU; Fraunhofer) Optimization of qubit-per-chip requirements; refinement based on feedback from WP4 (HEP validation) and WP5 (reliability testing); performance tuning for real-world data characteristics.					
T2.5	Library Integration (M25–M36; SNU; Naples; Fraunhofer) Integration of validated Multi-Chip Ensemble protocols into the unified "Physics-Aware QML" software library; documentation and API standardization.					
Deliverable		Month of delivery	Title of deliverable			
D2.1		M12	Report on Multi-Chip Protocols			
D2.2		M24	Software Module for Distributed Circuit Aggregation			

WP 3	Physics-Informed Optimization & Generative Models			Start month: 1	End month: 36
Contribution of project partners					
Partner number	1. SNU	2. Yonsei	3. Fraunhofer	4. Naples	
Total effort per partner (person*months)	36	2	4	24	
Aim of the WP To solve the "Trainability" and "Noise" bottlenecks by developing the Quantum Forward-Forward algorithm (QFF) driven by Hybrid Quantum Genetic Algorithms (HQGA), Quantum State Space Models (Q-SSM) for temporal expressibility, and Fuzzy-controlled Quantum Diffusion models. This WP addresses Objective 2 (Trainability) , Objective 3 (Temporal Expressibility) , and Objective 4 (Reliability) .					
Tasks					
T3.1	QFF-HQGA Protocol Definition (M1–M12; Naples; SNU) Development of Quantum Forward-Forward (QFF) algorithm with local "goodness" objectives; integration with Hybrid Quantum Genetic Algorithms (HQGA) for gradient-free optimization; benchmarking convergence against Backpropagation in Barren Plateau landscapes.				
T3.2	Q-SSM Architecture Design (M7–M18; SNU; Naples) Development of Quantum State Space Models (Q-SSM) integrating variational quantum circuits with classical LSTM-style gating mechanisms; implementation of three-branch quantum superposition for measurement-based interference effects; design of chunked recurrence processing for sequential data.				
T3.3	Fuzzy Quantum Diffusion Development (M13–M24; Naples; SNU) Design of Fuzzy Logic Controllers (FLC) to model NISQ noise as uncertainty; integration of FLC with Positive Operator-Valued Measures (POVMs) for continuous "unsharp" measurements; development of Hardware-Attention U-Net for reverse diffusion process.				
T3.4	Simulation Validation (M13–M30; Naples; SNU, Fraunhofer) Full implementation of QFF-HQGA, Q-SSM, and Fuzzy-Quantum Diffusion models; simulation-based validation of trainability improvements and noise resilience; comparison with classical baselines.				
T3.5	Library Integration (M25–M36; Naples; SNU, Fraunhofer) Integration of all optimization and generative model algorithms into the unified "Physics-Aware QML" library; API development and documentation.				
Deliverable		Month of delivery	Title of deliverable		
D3.1		M18	Algorithm Definition: QFF-HQGA, Q-SSM & Fuzzy-Diffusion		
D3.2		M30	Benchmarking Report: Trainability, Temporal Expressibility & Noise Resilience		

WP 4	High-Complexity Domain Validation (HEP, Neuro&Cyber)				Start month: 1	End month: 36
Contribution of project partners						
Partner number		1. SNU	2. Yonsei	3. Fraunhofer	4. Naples	
Total effort per partner (person*months)		42	24	6	6	
Aim of the WP To validate the foundational breakthroughs (WP2/WP3) using massive, complex datasets from High Energy Physics (LHC), Neuroscience (EEG, fMRI), and Cybersecurity (network intrusion detection), proving quantum advantage in scalability, temporal modeling, temporal modeling, and certified robustness. This WP addresses Objective 5 (Validation) .						
Tasks						
T4.1	Dataset Curation (M1–M12; Yonsei; SNU, Fraunhofer) Preparation of LHC data including CMS calorimeter images, high-level physics features, and detector waveform signals (Yonsei); curation of EEG/fMRI time-series datasets (SNU); preparation of network intrusion detection datasets with PII-screened logs (Fraunhofer); implementation of quantum feature maps for high-dimensional data (Fraunhofer).					
T4.2	Quantum Feature Mapping (M7–M18; Yonsei; SNU, Naples, Fraunhofer) Development of quantum encoding schemes for HEP, Neuroscience, and Cybersecurity data; optimization of feature map architectures for domain-specific characteristics; implementation of dimension reduction strategies compatible with Multi-Chip protocols.					
T4.3	HEP Validation (Multi-Chip + LHC) (M13–M30; Yonsei; SNU, Fraunhofer) Application of Multi-Chip Ensembles (from WP2) to LHC data classification using Quantum Vision Transformers; implementation of Q-ABCDiCo for background estimation; TCN-VQC for waveform denoising; benchmarking accuracy and generalization vs. single-chip and classical SOTA baselines.					
T4.4	Neuro Validation (Q-SSM + EEG/fMRI) (M13–M30; SNU; Naples, Fraunhofer) Application of Q-SSM (from WP3) to EEG/fMRI spatio-temporal analysis; implementation of QFF-HQGA driven QAOA/VQE for brain network modularity optimization; evaluation of long-range temporal dependency capture vs. classical transformers.					
T4.5	Cyber Validation (Fuzzy-Quantum Diffusion) (M13-M30; Fraunhofer; Naples, SNU) Application of Fuzzy-Quantum Diffusion (from WP3) to network anomaly detection via hardware fingerprinting; QUARK-certified robustness testing against adversarial attacks; benchmarking detection rates and false positive rates vs. classical intrusion detection systems.					
T4.6	Final Benchmarking vs SOTA (M25–M36; Yonsei; SNU, Naples, Fraunhofer) Comprehensive benchmarking of all quantum approaches against classical state-of-the-art across HEP, Neuro, and Cyber domains; statistical analysis of quantum advantage claims; preparation of validation reports.					
Deliverable		Month of delivery	Title of deliverable			
D4.1		M6	Curated "Challenge Datasets" (LHC, EEG/fMRI, Network logs)			
D4.2		M36	Validation Report: Roadmap to Quantum Advantage in HEP, Neuro & Cyber			

WP 5	Reliability, Safety & Certification				Start month: 1	End month: 36
Contribution of project partners						
Partner number		1. SNU	2. Yonsei	3. Fraunhofer	4. Naples	
Total effort per partner (person*months)		12	3	15	9	
Aim of the WP						
To define the first "QML Reliability Standard" and benchmark and certify the robustness of the developed QML models against cybersecurity threats, adversarial attacks, and privacy risks. This WP addresses Objective 4 (Reliability) certification aspects.						
Tasks						
T5.1	Requirements Specification (M1–M12; Fraunhofer; All)					
	Definition and development of reliability and safety requirements for QML models; specification of robustness and reliability metrics (e.g., Lipschitz continuity, noise stability, data leakage); identification of cybersecurity vulnerability and inherent defenses relevant to model types developed in WP2 and WP3.					
T5.2	Robustness Enhancement (M7–M18; Fraunhofer; Naples)					
	Development of cybersecurity defense mechanisms to improve model robustness, via architectural changes (e.g., robust feature maps, in cooperation with T4.2) and additional techniques (e.g., adversarial training, differential privacy).					
T5.3	Vulnerabilities and Defenses Testing (M13–M30; Fraunhofer; Naples)					
	Application of Evolutionary QML to Network Intrusion Detection; stress-testing the Fuzzy-Quantum Diffusion models with gradient-based adversarial attacks; adversarial certification for recurrent (Q-SSM) and attention-based models.					
T5.4	Certification Protocol Development (M25–M36; Fraunhofer)					
	Synthesis of results into a formal certification protocol for industrial QML applications; development of certification guidelines for safety-critical deployment; documentation of reliability standards.					
Deliverable		Month of delivery		Title of deliverable		
D5.1		M18		Robustness Report: Metrics & Initial Assessment		
D5.2		M36		QML Reliability Standard & Certification Protocols		

WP 6	Dissemination, Exploitation & Communication				Start month: 1	End month: 36
Contribution of project partners						
Partner number		1. SNU	2. Yonsei	3. Fraunhofer	4. Naples	
Total effort per partner (person*months)		12	4	5	5	
Aim of the WP						
To maximize impact through Open Science publications, release of the "Physics-Aware QML" library, and engagement with the industrial and public sectors.						
Tasks						
T6.1		Communication Planning (M1–M12; Fraunhofer; SNU) Development of project website and communication strategy; establishment of social media presence; creation of project visual identity and materials.				
T6.2		Publications & Conferences (M13–M36; All) Publication of results in Open Access journals (Nature Comms, PRX Quantum, Quantum Machine Intelligence, etc.); management of the GitHub repository and Zenodo data archives; presentations at major conferences (IEEE QCE, CHEP, etc.).				
T6.3		Hackathon & Summer School (M25–M30; Fraunhofer; SNU) Organization of the "Quantum Robustness Hackathon" (M30) challenging participants to attack/defend QML models; summer school on Physics-Aware QML for early-career researchers.				
T6.4		Open-Source Library Release (M31–M36; SNU; Naples, Fraunhofer) Final release of the integrated "Physics-Aware QML" open-source library under Apache 2.0 license; comprehensive documentation and Jupyter notebook tutorials; Zenodo archival with DOI assignment.				
Deliverable		Month of delivery	Title of deliverable			
D6.1		M36	Open Source "Physics-Aware QML" Library Release			
D6.2		M36	Final Plan for Exploitation and Standardization			

Work package overview (total effort per WP and partner in person.months)

Partner	WP1	WP2	WP3	WP4	WP5	WP6	Total
1. SNU	9	60	36	42	12	12	171
2. Yonsei	1	6	2	24	3	4	40
3. Fraunhofer	2	4	4	6	15	5	36
4. Naples	1	9	24	6	9	5	54
Total	13	79	66	78	39	26	301

3.3 Management structure, milestones, risk assessment

Organizational Structure and Decision-Making

Project Coordinator (PC): Prof. Jiook Cha (SNU) is the single point of contact for the QuantERA Secretariat. The PC is responsible for overall progress monitoring, submission of deliverables, and conflict resolution.

General Assembly (GA): The ultimate decision-making body, composed of one representative from each partner. The GA meets bi-annually to review progress against Milestones. Decisions are made by consensus; if consensus fails, a majority vote (one vote per partner) applies.

Scientific Steering Committee (SSC): Composed of the Work Package Leaders (WP1/WP2: SNU; WP3: Naples; WP4: Yonsei; WP5/WP6: Fraunhofer). The SSC meets quarterly to ensure technical alignment across WPs.

Innovation & Ethics Board: This board oversees IPR management (led by Fraunhofer IKS), Open Science compliance (led by Fraunhofer IKS), and ethical handling of sensitive data. **SNU leads medical/neuroscience ethics**** (EEG/fMRI data, IRB protocols, GDPR/PIPA compliance); **Fraunhofer handles Cybersecurity data ethics** (PII screening for network logs).

Appropriateness of Structure

This structure is appropriate because it is **lean** (minimizing administrative overhead for a 4-partner consortium) yet **robust** (ensuring all scientific disciplines—Physics, Computer Science, Engineering—have equal representation in the GA). The geographic distribution is balanced with clear communication channels to bridge the Asia-Europe collaboration.

List of Milestones

Milestone	Month	WP Involved	Title
MS1	M12	WP2, WP3, WP4, WP5	Theoretical Foundations Established
MS2	M24	WP2, WP3, WP4	Algorithmic Validation (Simulation)
MS3	M30	WP4, WP5	Cross-Domain Integration
MS4	M36	WP2, WP3, WP5, WP6	Final Deliverables

Success Criteria

- **MS1:** Multi-Chip Ensemble protocol specifications complete; QFF-HQGA and Q-SSM architectures defined; Fuzzy Logic Controllers specified; reliability metrics defined; Domain datasets (LHC, EEG/fMRI, Network Logs) preprocessed and feature-mapped
- **MS2:** QFF-HQGA converges on >10-layer circuits where backpropagation fails; Q-SSM achieves $\geq 1.5\times$ classical memory capacity; Fuzzy-Quantum Diffusion achieves FID $\leq 1.5\times$ classical diffusion baseline
- **MS3:** Multi-Chip Ensemble reconstructs LHC particle correlations across 2+ parallel chips; Q-SSM demonstrates superior long-range temporal capture on EEG/fMRI; QML models pass robustness certification against adversarial attacks
- **MS4:** Open-source "Physics-Aware QML" library released (Apache 2.0); Final benchmarks published demonstrating advantage in scalability (O1), trainability (O2), temporal expressibility (O3), and reliability (O4: Lipschitz $L < 2.0$ certification); Certification Standard documented

Critical Risks for Implementation

Description of Risk	WPs	Proposed Risk-Mitigation Measures
Scientific: Barren Plateaus persist despite QFF-HQGA methods, preventing convergence in deep circuits.	WP3, WP4	Mitigation: We do not rely on a single optimization strategy. The HQGA component provides gradient-free global search via entangled crossover and quantum elitism. If QFF proves unstable, HQGA serves as a robust fallback that navigates the optimization landscape without gradient dependence. Additionally, the local "goodness" objectives of QFF inherently reduce gradient variance.
Technical: Ensemble Aggregation fails to achieve accuracy targets due to weak individual learners on small chips.	WP2, WP4	Mitigation: We will implement advanced "Weighted Voting" mechanisms where each chip's output weight is determined by its reliability score. The Quantum Transformer ansatz is specifically designed to maximize expressibility even on small chips. Selective Entanglement between chips processing globally dependent features will enhance collective expressivity.
Technical: Q-SSM fails to capture long-range temporal correlations in EEG/fMRI data, underperforming classical baselines.	WP3, WP4	Mitigation: The three-branch quantum superposition ($\alpha, \beta, \gamma \in \mathbb{C}$) architecture provides interference-like effects while LSTM-style gating handles sequential memory. If temporal modeling is insufficient, we will enhance the bidirectional processing layer and increase chunk overlap. Alternative QSVT-enhanced attention mechanisms provide a fallback architecture.
Technical: Fuzzy-Quantum Diffusion fails to achieve target generative fidelity under NISQ noise conditions.	WP3, WP4	Mitigation: The Fuzzy Logic Controllers using POVMs (Positive Operator-Valued Measures) are specifically designed to model noise as continuous "degrees of uncertainty" rather than binary errors. If the Hardware-Attention U-Net underperforms, we can pivot to Quantum GANs enhanced with the same Fuzzy Controllers to stabilize adversarial training dynamics.
Operational: Delays in accessing sensitive domain data (LHC physics data, EEG/fMRI clinical data, network security logs).	WP4, WP5	Mitigation: All partners have established direct access: Prof. Yoo is a CMS Collaboration member with CERN LHC access; SNU has IRB-approved neuroimaging protocols. For contingency, open-source proxy datasets (CERN Open Data Portal, PhysioNet EEG archives, KDD Cup 99) are identified to proceed with algorithmic development without delay.
Management: IPR disputes regarding industrial exploitation of jointly-developed algorithms.	WP1, WP6	Mitigation: A comprehensive Consortium Agreement (CA) based on the DESCA model will be signed before project start, clearly defining background/foreground IP ownership and licensing terms. The "Open Source by Default" strategy (Apache 2.0) for the core library minimizes commercial disputes while preserving academic recognition.

3.4 Consortium as a whole

Matching Objectives and Expertise

The consortium is designed as a **trans-continental feedback loop** bridging Foundational Quantum Theory (O1–O4) and Rigorous Domain Validation (O5). No single partner possesses the full spectrum of skills required to deliver the "Physics-Aware QML Stack," necessitating this specific four-partner collaboration:

Role	Partner	Objective Addressed	Key Contribution
Architects	SNU (Korea)	O1: Scalability	Multi-Chip Ensemble protocols, Q-SSM, Quantum Transformers
Optimizers	Naples (Italy)	O2: Trainability, O4: Reliability	QFF-HQGA, Fuzzy Quantum Diffusion, POVMs
Validators	Yonsei (Korea)	O5: Domain Validation	LHC/CMS data, Q-ABCDISCo, TCN-VQC, Quantum ViT
Certifiers	Fraunhofer IKS (Germany)	O4: Reliability Certification	Robust QML, Lipschitz bounds, adversarial testing, benchmarking

Complementarity and Contribution

The partners complement each other through a "**Scale-Logic-Trust**" triangle with domain grounding:

- **SNU (Scale)** Contributes **171 PM** (57% of total effort), leading WP1/WP2. Provides scalable architectures (Multi-Chip Ensembles with Selective Entanglement, variance reduction $\propto 1/k$) essential for fragmented NISQ hardware.
- **Naples (Logic)** Contributes **54 PM** leading WP3. Without Naples, deep architectures would be untrainable (Barren Plateaus). QFF-HQGA (entangled crossover, quantum elitism) and Fuzzy Logic Controllers using POVMs provide the resilience layer for hardware-native generative modeling.
- **Fraunhofer IKS (Trust)** Contributes **36 PM** leading WP5/WP6. Transforms outputs into certified software assets (Lipschitz $L < 2.0$ guarantees) via robustness analyses, enabling safety-critical deployment.
- **Yonsei (Physical Reality)** Contributes **40 PM** leading WP4. CMS Collaboration membership provides LHC petabyte-scale data with intrinsic non-local correlations—the ideal testbed proving Multi-Chip Ensembles preserve quantum expressivity.

Resource Adequacy: The **301 Person-Months** ensure adequate capacity: SNU's large team handles coordination (WP1) and architecture development (WP2); Naples provides specialized optimization expertise; Fraunhofer brings industrial certification experience; Yonsei manages the HEP data pipeline.

Industrial/Commercial Involvement

Fraunhofer IKS bridges foundational research to industry. As a leading German RTO coordinating one of the **Munich Quantum Valley** consortia and European QC Benchmarking Committee, Fraunhofer provides direct channels to automotive, aerospace, and medical technology sectors.

This industrial involvement is consistent with Section 2 exploitation measures: The "**QML Reliability Standard**" (WP5, D5.2) validates commercial viability of "Trustworthy QML" services with formal metrics (gradient variance, entangling capability, Lipschitz bounds). The "**Open Source by Default**" strategy (Apache 2.0) maximizes academic impact while the certification layer creates commercial value through consulting services for companies requiring safety guarantees before QML deployment.

3.5 Description of the consortium

Jiook Cha Project Coordinator	Seoul National University / Department of Psychology, Department of Brain and Cognitive Sciences, Interdisciplinary Program in Artificial Intelligence
Expertise: Seoul National University is South Korea's premier research university. Prof. Jiook Cha's lab specializes in Quantum Machine Learning (QML) architectures and their application to Foundation Models for Neuroscience. The group possesses unique expertise in: 1. Multi-Chip Ensembles: Developing scalable approaches to train QML models across multiple QPUs, with mathematical framework demonstrating variance reduction $\propto 1/k$. Applied to image data, EEG/fMRI, and Reinforcement Learning. 2. Quantum Time-Series Analysis: Pioneering quantum algorithms for temporal data, including Quantum Transformers with Quantum Singular Value Transformation (QSVT) achieving polylogarithmic complexity (IEEE QCE 2025, QML Best Paper Award) for fMRI/EEG analysis. 3. QML for Neuroscience: Bridging quantum computing with brain science through specialized neural network architectures including Quantum Time-series Transformers and Hybrid Quantum Temporal Convolutional Networks. Principal Investigator: Prof. Jiook Cha (Male) Prof. Cha is a leading expert in applied QML architectures and computational neuroscience (h-index: 29; 3,100+ citations). His research focuses on creating scalable quantum algorithms for high-dimensional biological data, with particular emphasis on overcoming NISQ hardware limitations through distributed computing strategies. He has extensive experience with large-scale neuroimaging datasets including ABCD Study, UK Biobank, and Human Connectome Project. His lab maintains productive collaborations with Brookhaven National Laboratory on QML research. Relevant Achievements: • IEEE QCE 2025 QML Best Paper Award for "Quantum Time-series Transformer" applying QSVT-enhanced attention to fMRI analysis with ADHD biomarker identification. • Developing multi-chip ensemble approaches with Selective Entanglement, addressing scalability, trainability, and noise resilience challenges in QML. • NeuroMamba (NeurIPS 2025): First state-space foundation model for full-resolution 4D fMRI with linear-time complexity and SOTA performance on HCP. • Scalable Diffusion Transformer (NeurIPS 2025): First diffusion transformer for voxelwise 4D fMRI conditional generation (0.83 task-evoked map correlation). • DIVER-0 (ICML 2025): Channel equivariant EEG foundation model achieving competitive results with only 10% of typical pretraining data.	
Role in project: Coordinator & QML Architecture Lead. SNU leads WP1 (Management & Coordination) and WP2 (Scalable Distributed Quantum Architectures) by implementing the Multi-Chip Ensemble protocols and Q-SSM architectures. Prof. Cha also drives the theoretical development of Neuro-Generative validation in WP4, utilizing his expertise in Quantum Transformers and foundation models to process EEG/fMRI data. SNU is responsible for medical/neuroscience data ethics, including IRB protocols and GDPR/PIPA compliance for EEG/fMRI datasets. Total effort: 171 PM.	

Hwidong Yoo	Yonsei University / Department of Physics
Expertise: Yonsei University is a leading private research university in Korea with a strong legacy in High Energy Physics (HEP). Prof. Hwidong Yoo is a key member of the CMS Collaboration at CERN, providing the consortium with direct access to massive-scale "Big Science" data. Yonsei hosts IBM Quantum System One (127-qubit), deployed November 2024—the second such system on a university campus worldwide. 1. High Energy Physics: Deep expertise in LHC particle jet analysis, tracking, event classification, and detector physics. Key contributor to the Snowmass 2021 White Paper on dual-readout calorimetry shaping future detector R&D. 2. Dual-Readout Calorimetry: Pioneering work on 3D shower shape reconstruction using SiPM timing, producing 5-dimensional spatio-temporal tensors (x, y, z, E, t)—the exact data structure motivating the Quantum Time-series Transformer. 3. Scalable Computing: Experience scaling deep learning on HPC infrastructure (Nurion supercomputer at KISTI) across thousands of parallel processors, directly applicable to distributed quantum system coordination. 4. Quantum for HEP: Experience in benchmarking quantum algorithms (Quantum Vision Transformers, Q-ABCDiCo, TCN-VQC) against classical Deep Learning for particle physics tasks. Principal Investigator: Prof. Hwidong Yoo (Male) Prof. Yoo is an expert in experimental high-energy physics and the application of advanced computing to particle detection. He leads the validation efforts, bridging the gap between abstract quantum algorithms and physical reality. Relevant Achievements: • Leadership roles in the CMS Collaboration (CERN) for particle data analysis; key author of Snowmass 2021 dual-readout calorimetry white paper. • Development of classical baselines for CMS calorimeter image, high-level feature, and waveform analysis tasks. • Large-scale distributed training of CNNs on Nurion HPC (1.85× signal efficiency improvement over cut-based methods). • Pioneering work in applying Quantum Vision Transformers, Quantum Neural Networks (Q-ABCDiCo), and hybrid TCN-VQC models to CMS classification, background estimation, and waveform denoising.	
Role in project: Domain Validation Lead (HEP). Yonsei leads WP4 (High-Complexity Domain Validation). They curate LHC datasets (CMS calorimeter images, high-level physics features, detector waveforms), implement quantum feature maps for high-dimensional particle data, and rigorously benchmark Multi-Chip Ensemble protocols (WP2) against classical SOTA. Prof. Yoo's HPC scaling experience directly supports WP2 (Scalable Distributed Quantum Architectures), while Yonsei's IBM Quantum System One provides critical hardware validation infrastructure. Total effort: 40 PM.	

Jeanette Miriam Lorenz	Fraunhofer Institute for Cognitive Systems IKS / Department "Quantum-enhanced AI"
Expertise: Fraunhofer IKS is Germany's leading applied research institute for Safe and Reliable Software Engineering and Artificial Intelligence ("Safe Intelligence"). PD Dr. habil. Jeanette Miriam Lorenz leads the QuaST project (quantum-enabling services and tools for industrial applications) and is affiliated with the Munich Quantum Valley (MQV) and Bavarian Competence Center for Quantum Security and Data Science (BayQS). Her group specializes in Reliable Quantum Computing for Applications, uniquely combining quantum generalization theory with industrial-scale benchmarking frameworks. - Quantum Generalization Theory: Pioneering generalization bounds for hybrid quantum-classical models (2025), proving distributed architectures maintain performance—crucial for validating Multi-Chip Ensembles (WP2). - Quantum Randomized Smoothing: Demonstrated quadratic speedups in robustness certification using Quantum Amplitude Estimation (2024), applicable to Prof. Yoo's HEP 5D tensor data where classical certification is prohibitive. - Noise-as-Resource Theory: Theoretical grounding (2024) for repurposing quantum noise in diffusion processes—directly informing Fuzzy-Quantum Diffusion collaboration with Naples (WP3). - Safety & Security Analysis: Developer of tools and methodologies for QML robustness certification, including Lipschitz bound estimation and adversarial testing protocols for safety-critical applications (WP5). Principal Investigator: PD Dr. habil. Jeanette Miriam Lorenz (Female) PD Dr. Lorenz heads the department 'Quantum-enhanced AI' at Fraunhofer IKS and is associate professor at LMU Munich and brings extensive ATLAS experiment (CERN LHC) experience in supersymmetry searches, dark matter, and Higgs physics. Her research establishes rigorous theoretical guarantees of QML performance beyond heuristic claims. Relevant Achievements: ● Generalization bounds for hybrid QML (2025), proving shallow evolutionary-trained circuits match deeper gradient-optimized circuits. ● Quantum Randomized Smoothing achieving quadratic certification speedups via Grover-based amplitude estimation (2024). ● Quantum-classical CNNs for medical imaging (IEEE QCE, 2022) showing quantum models outperform classical baselines on breast tumor recognition, particularly with limited training data. ● Coordination of the European QC Benchmarking Coordination Committee (EQCBC).	
Role in project: Reliability & Certification Lead Fraunhofer leads WP5 (Reliability, Safety & Certification) and WP6 (Dissemination, Exploitation & Communication). Dr. Lorenz provides QML Reliability Standard ensuring unified safety and security analysis across all partners. Her generalization theory validates Multi-Chip Ensembles (WP2); noise-as-resource research informs Fuzzy-Quantum Diffusion (WP3); Quantum Randomized Smoothing certifies Q-SSM and attention models for HEP time-series (WP4). Fraunhofer leads the Cybersecurity domain validation (T4.5), applying Fuzzy-Quantum Diffusion and adversarial robustness testing to network intrusion detection. Her ATLAS/CERN statistical expertise supports HEP and Neuro validation. She coordinates the "Quantum Robustness Hackathon" and Open Science compliance. Total effort: 36 PM.	

Giovanni Acampora	University of Naples Federico II / Department of Physics “Ettore Pancini”
Expertise: The University of Naples Federico II is one of the oldest universities in the world. Prof. Giovanni Acampora's QUASAR Lab (Quantum Computing and Smart Systems Laboratory) is globally recognized for Computational Intelligence and Fuzzy Logic. They bring the critical "classical intelligence" layer to the quantum stack. 1. Fuzzy Logic Systems: World-class expertise in Type-2 Fuzzy Sets and Fuzzy Logic Controllers, essential for modeling NISQ noise as continuous uncertainty. 2. Evolutionary Computation: Deep experience in Hybrid Quantum Genetic Algorithms (HQGA) and bio-inspired optimization to solve non-convex problems; developer of EVOVAQ, the first open-source evolutionary algorithm toolbox for variational quantum circuits [17]. 3. Quantum Computational Intelligence: Pioneering the fusion of Fuzzy Logic with Quantum Computing, including Positive Operator-Valued Measures (POVMs) for "unsharp" quantum measurements. 4. Distributed Quantum Systems: Demonstrated quantum fuzzy inference engines on the D-NISQ model, proving divide-and-conquer approaches viable for multi-chip quantum architectures. Principal Investigator: Prof. Giovanni Acampora (Male) Prof. Acampora is a Senior Member of IEEE, Editor-in-Chief of Springer Quantum Machine Intelligence, and a pioneer in Quantum Computational Intelligence. His work focuses on hybridizing soft computing with quantum mechanics to solve trainability and noise-resilience issues. Relevant Achievements: • Definition of Quantum Fuzzy Logic frameworks for uncertainty management using POVMs. • Development of Hybrid Quantum Genetic Algorithms (HQGA) with entangled crossover and quantum elitism for optimizing Variational Quantum Circuits. • Application of Quantum Forward-Forward (QFF) algorithms with local "goodness" objectives to prevent Barren Plateaus in QML. • Hardware validation of quantum fuzzy inference engines on both superconducting and ion-trap backends, demonstrating real-world implementability (IEEE TQE, 2024). • Analysis of quantum noise effects on evolutionary algorithms, providing theoretical grounding for the noise-as-resource paradigm. Role in project: Optimization & Intelligence Lead Naples leads WP3 (Physics-Informed Optimization & Generative Models). They provide the QFF-HQGA algorithms to drive gradient-free training, design the Fuzzy Logic Controllers for the Fuzzy-Quantum Diffusion models using POVMs, and develop Hardware-Attention U-Net architectures for noise-resilient generative modeling. Prof. Acampora's EVOVAQ framework provides the software infrastructure for WP2 (Scalable Distributed Quantum Architectures), while his D-NISQ expertise validates the divide-and-conquer approach. Naples also supports WP4 (Neuro Validation) through QFF-HQGA driven brain connectivity optimization, and WP5 in adversarial evolutionary testing. Total effort: 54 PM.	

3.6 Consortium agreement principles (partner's rights and duties, IPR management, data management, open access, GDPR)

Consortium Agreement & Decision Making

A **Consortium Agreement (CA)** based on the **DESCA model** will be signed prior to project start. SNU (Coordinator) manages the QuantERA interface; the **General Assembly (GA)** decides by consensus (majority vote if needed); the **Scientific Steering Committee (SSC)** ensures quarterly technical alignment.

Intellectual Property Rights (IPR) Management

- **Background IP:** Partners retain ownership; royalty-free access for project purposes.
- **Foreground IP:** Co-owned by contributors; "**Open Source by Default**" under **Apache 2.0**.
- **Commercial Exploitation:** Fraunhofer leads via Innovation Board; certification tools may be licensed commercially.

Data Management & Open Access

Following **FAIR principles**: 100% Open Access publications (arXiv/Zenodo); benchmark datasets on **Zenodo** with DOIs; code on **GitHub**; models on **HuggingFace**. DMP delivered M3 (D1.1).

GDPR & Ethical Compliance

- **Neuroscience Data:** Pseudonymized at SNU; GDPR/PIPA compliant; only anonymized derivatives shared.
- **Cybersecurity Data:** PII-screened before sharing; used for robustness testing (WP5).
- **Ethics Oversight:** SNU leads medical ethics (IRB, EEG/fMRI); Fraunhofer leads IPR/Open Science and Cybersecurity data ethics.

3.7 Significant facilities and large equipment available to the consortium to perform the project

(max. ½ page)

1. High-Performance Computing & Simulation

SNU operates GPU-accelerated HPC environments (PennyLane Lightning, CUDA-Q) capable of simulating 30+ qubits, essential for Multi-Chip Ensemble validation. Naples contributes the QUASAR Lab infrastructure and **EVOVAQ** [17]—the open-source evolutionary toolbox serving as the software backbone for QFF-HQGA optimization (WP3).

2. "Big Science" Data Infrastructure

Through Prof. Yoo's CMS Collaboration membership, Yonsei provides access to petabyte-scale **CERN LHC data** (calorimeter images, waveforms). These datasets contain intrinsic non-local correlations ideal for validating distributed quantum protocols. Yonsei also leverages the **KISTI Nurion supercomputer** for large-scale distributed training tasks.

3. Quantum Hardware Access

Yonsei hosts the **IBM Quantum System One** (127-qubit, deployed Nov 2024), providing on-site access to superconducting hardware for final algorithm validation. Fraunhofer IKS secures access to diverse backends (superconducting, ion-trap) via **Munich Quantum Valley (MQV)**, enabling multi-platform robustness certification (WP5).

4. Robustness Certification Infrastructure

Fraunhofer IKS contributes proprietary tools for QML certification, including Lipschitz bound estimation, adversarial testing, and Quantum Randomized Smoothing. Developed through QuaST and European QC Benchmarking coordination, these tools form the core reliability infrastructure for WP5.

3.8 Link with ongoing projects

None of the consortium partners currently have ongoing projects directly linked to the specific topics or objectives of this proposal. Consequently, there are no overlapping funding sources to declare for any partner.

3.9 Financial plan

The total requested budget for the consortium is **€1,065,945.34**, distributed to ensure the successful execution of the "Physics-Aware QML" work program.

1. Personnel Costs (€569,211.76): This is the largest budget category, reflecting the labor-intensive nature of foundational algorithmic research.

- **Partner 1 (SNU, €163,905.88):** Full-time PhD students dedicated to developing Multi-Chip Ensemble protocols (WP2) and Quantum Transformers (WP4).
- **Partner 2 (Yonsei, €84,705.88):** Three graduate students to handle massive data preprocessing and benchmarking of LHC particle jets (WP4).
- **Partner 3 (Naples, €56,000.00):** A Post-doc researcher specializing in Evolutionary Algorithms and Fuzzy Logic to lead optimization tasks in WP3.
- **Partner 4 (Fraunhofer IKS, €264,600.00):** A senior researcher/Post-doc with industrial experience to drive Certification and Reliability Standards (WP5).

2. Equipment (€71,098.04): Essential hardware is required for simulating quantum circuits and training hybrid models.

- **Partner 1 (SNU, €16,833.33) & Partner 3 (Naples, €12,500.00):** High-performance GPU workstations critical for simulating the Quantum Forward-Forward (QFF) algorithm and its intensive matrix operations (WP3).
- **Partner 2 (Yonsei, €11,764.71):** Servers to store and process petabyte-scale LHC datasets locally before quantum feature mapping (WP4).
- **Partner 4 (Fraunhofer IKS, €30,000.00):** Dedicated budget for "Access to Quantum Computers" to purchase compute time on cloud-based NISQ backends (e.g., IBM Quantum, IonQ) for validating Robustness metrics (WP5).

3. Travel (€162,122.36): Travel costs are strictly necessary for ensuring cohesion between the Asian and European partners.

- **Partner 1 (SNU, €69,534.12) & Partner 2 (Yonsei, €70,588.24):** Reflects long-distance travel to Europe for bi-annual consortium meetings and "Methodology Swap" workshops (WP6), plus student travel to major conferences.

- **Partner 3 (Naples, €7,000.00) & Partner 4 (Fraunhofer IKS, €15,000.00):** Intra-European travel for meetings, the final "Quantum Robustness Hackathon" in Munich, and international dissemination.

4. Other Goods, Services & Consumables (€56,104.71):

- **Partner 1 (SNU, €32,781.18):** Allocated for management, workshop organization costs, and dissemination activities (WP1).
- **Partner 2 (Yonsei, €18,823.53 total):** Consumables for data storage media (€11,764.71) and other services for HEP validation (€7,058.82).
- **Partner 3 (Naples, €4,500.00 total):** Gold Open Access publication fees (€3,500.00) and minor consumables (€1,000.00).

5. Overheads (€207,408.47): Calculated according to the specific national rules of each funding agency (NRF Korea, MUR Italy, DFG Germany).

Summary of Resource Commitment: The allocation ensures SNU and Yonsei have the personnel and travel resources to bridge the geographical gap and drive the heavy theoretical/experimental load, while Fraunhofer IKS is resourced to access real quantum hardware for industrial-grade certification. University of Naples is efficiently budgeted to provide specialized optimization logic.

4. Ethical issues

Foreseeable Issues & Mitigation Strategies. This project addresses two sensitive domains through a "Safety-by-Design" framework overseen by the **Innovation & Ethics Board** (Lead: Fraunhofer IKS):

- **Human Biological Data (Neuroscience):** WP4 processes EEG/fMRI data, raising privacy and bias risks.
 - *Legal Compliance:* We strictly adhere to **GDPR** (EU) and **PIPA** (South Korea). SNU pseudonymizes all clinical data at the source before consortium sharing.
 - *Technical Mitigation:* A core objective is developing **Fuzzy-Quantum Diffusion (WP3)** to generate high-fidelity synthetic datasets. This minimizes the exposure of real patient data while maintaining statistical validity.
 - *Bias Mitigation:* To prevent algorithmic discrimination in brain signal analysis, we implement **gender-stratified validation** (balanced 50/50 male/female training data) as defined in Section 1.5.
- **Dual-Use Potential (Cybersecurity):** WP5 develops adversarial attack protocols to stress-test QML models.
 - *Responsible Disclosure:* We adopt an "As Closed As Necessary" policy. Detailed attack vectors are encrypted and restricted to the consortium. Public dissemination is limited to the **Defense/Certification Standards** rather than offensive tools.

Justification of Methodology. The use of high-dimensional human data is scientifically unavoidable to validate the **Temporal Expressibility of Q-SSM (Objective 3)**; synthetic toy datasets fail to capture the long-range correlations required to prove quantum advantage. Similarly, developing rigorous adversarial attacks is the only mathematical method to certify **Lipschitz bounds (Objective 4)**, converting experimental QML into safe, industrial-grade technology.

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